



IDBI INNOVATE

2026

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Team Details

- a. **Team name:** Lambdac Computing
- b. **Team leader name:** Reshma Mohan
- c. **Problem Statement:** Problem Statement 3 - Financial Health Score

Brief about the idea

CreditPulse - Financial Health Score : an explainable MSME Financial Health Card built from alternate data

- The problem: New-to-Credit and New-to-Bank MSMEs lack audited financials, so viable borrowers get rejected. Rich alternate data (GST, UPI, AA, EPFO) exists, but there is no unified assessment framework.
- The idea: every operating business leaves an electronic trail across independently governed systems - a tax authority, a regulated bank, a labour-welfare body, a toll network, a state utility. CreditPulse fuses 25 such sources into signals no single document can fake.
- The output: a multidimensional health score across 5 dimensions, an explainable grade of 1 to 10, a lending recommendation with an indicative limit, and plain-language reason codes - in near real time.
- How it's built: rules first, machine learning second. Every score carries its reasons, and the model never improves a score as a risk factor worsens (enforced and tested).

Note: this entry is one track of CreditPulse, one codebase and one deployment answering three problem statements (PS3 Financial Health Score, PS4 Early Warning / Default Prediction, PS5 Fraud Intelligence).

Opportunities

How different is it from any of the other existing ideas?

- Most scoring solutions ship a single black-box number. CreditPulse shows its work: 5 dimension scores, reason codes on every card, and a cross-source Turnover-Authenticity check (GST vs bank vs UPI vs e-way) that catches inflated declared turnover no single document reveals.
- Against incumbents (Perfios, CRIF, FinBox): focused on NTC/NTB onboarding, native to Account Aggregator and ULI, and full explainability - deployable under RBI digital-lending and model-risk expectations.

How will it be able to solve the problem?

- Makes credit-invisible MSMEs assessable: rejected-but-viable borrowers become safe approvals, while inflated and stressed cases get flagged - onboarding grows and portfolio quality improves at the same time.
- A sub-minute first-pass assessment replaces weeks of document collection, and the officer keeps the final call with reasons on record.

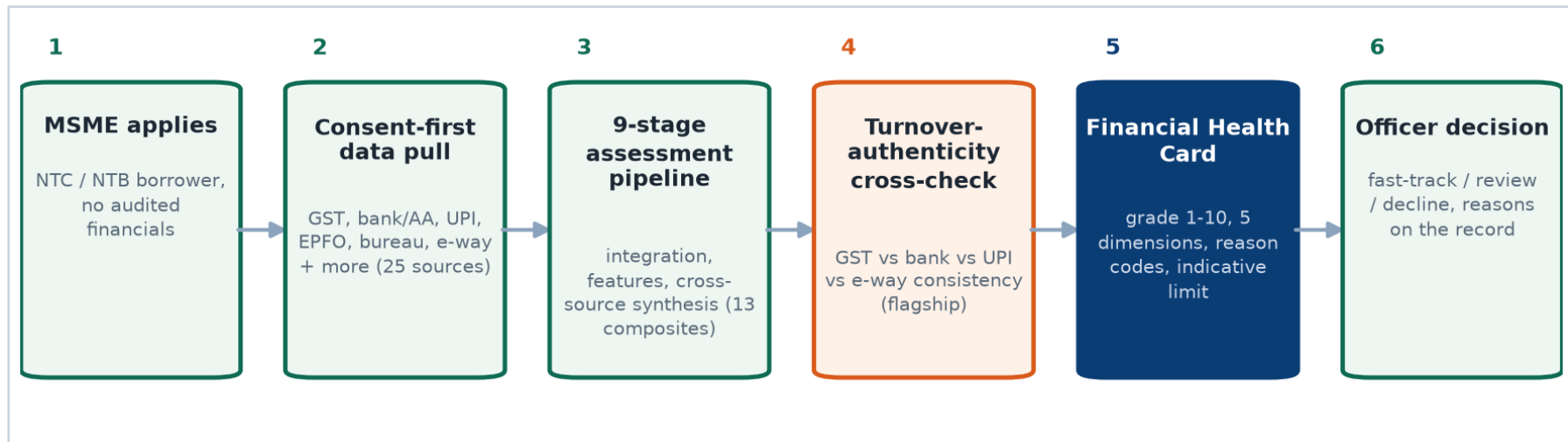
USP of the proposed solution

- Explainable and auditable throughout, consent-first on Account Aggregator data, and a working deployed prototype with a reproducible repository and eval harness.

List of features offered by the solution

- Explainable Financial Health Card: composite score, grade 1 to 10, five dimensions, recommendation and indicative limit
- 25 alternate-data sources fused into 93 features; 13 cross-source composite indicators, each with a manipulation-resistance rationale
- Turnover-Authenticity cross-check (key signal): GST vs bank vs UPI vs e-way consistency
- Interpretable WOE / IV scorecard plus a monotonicity-constrained LightGBM lift, with probability calibration
- Native reason codes plus SHAP; every card states its dominant drivers in plain language
- Peer segmentation (descriptive only) and a data-confidence score aware of source completeness
- Nine-stage transparent assessment pipeline, animated live in the product
- Dual-audience UI: Simple mode for business users, Technical mode for model-risk teams
- Eval harness with AUC, Gini, KS, calibration and PSI on a leakage-resistant holdout (one command: make eval)
- The officer stays in charge: recommendation, override and audit trail

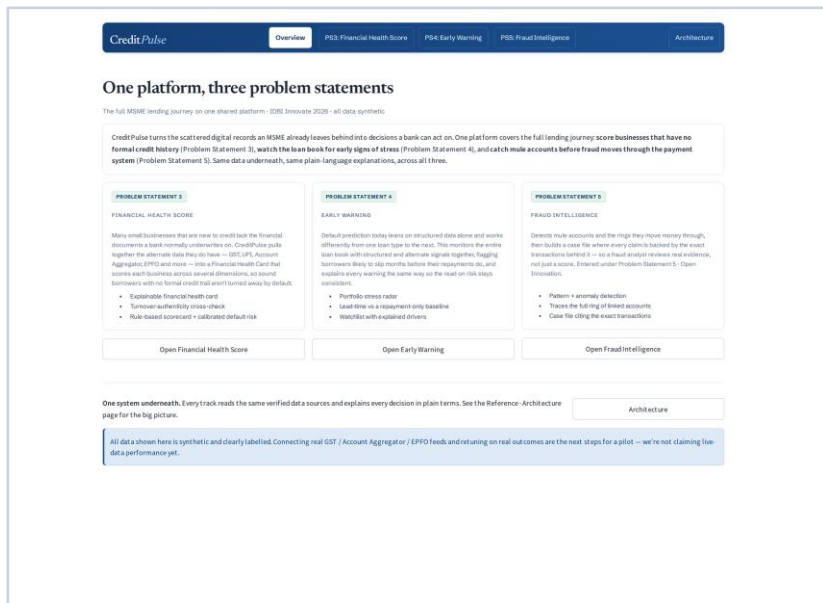
Process flow diagram or Use-case diagram



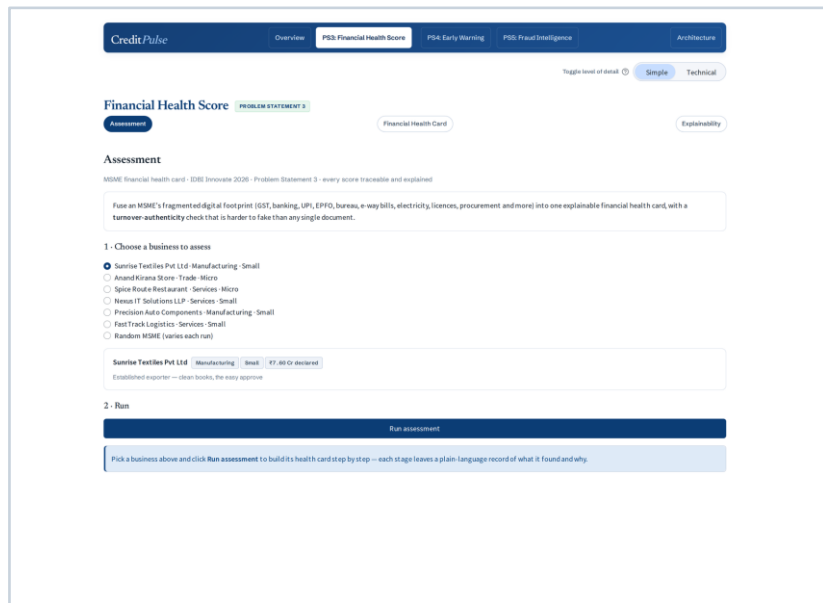
End-to-end use case: from a thin-file MSME application to an explainable decision. Every step is deterministic and reproducible; the pipeline is animated live in the prototype.

Wireframes/Mock diagrams of the proposed solution (optional)

The prototype is implemented and deployed; actual product screens below stand in for wireframes.

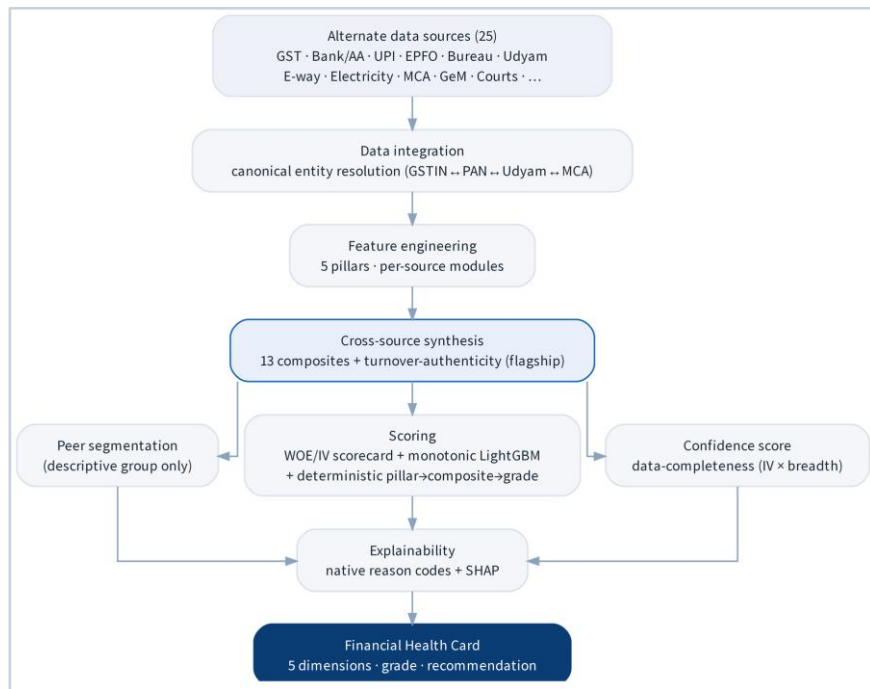


Platform landing page: one card per problem statement



Assessment: archetype picker and the nine-stage pipeline, animated live

Architecture diagram of the proposed solution



Rendered live inside the product (Architecture page, Problem Statement 3 view).

Module boundaries, enforced by tests

- frontend renders only; backend orchestrates and owns all narratives; ml computes (pure pandas / scikit-learn / LightGBM)

Deployment

- Stateless scoring; engines pre-fitted at build time, load instantly (version-guarded)
- One Docker container serves all three problem statements, with a deep link per track

Integration posture

- AA / ULI / OCEN-ready adapters; synthetic generators swap for sandbox APIs at stage 2
- CI regenerates the cohort, runs 296 tests and the eval harness on every push

Technologies to be used in the solution

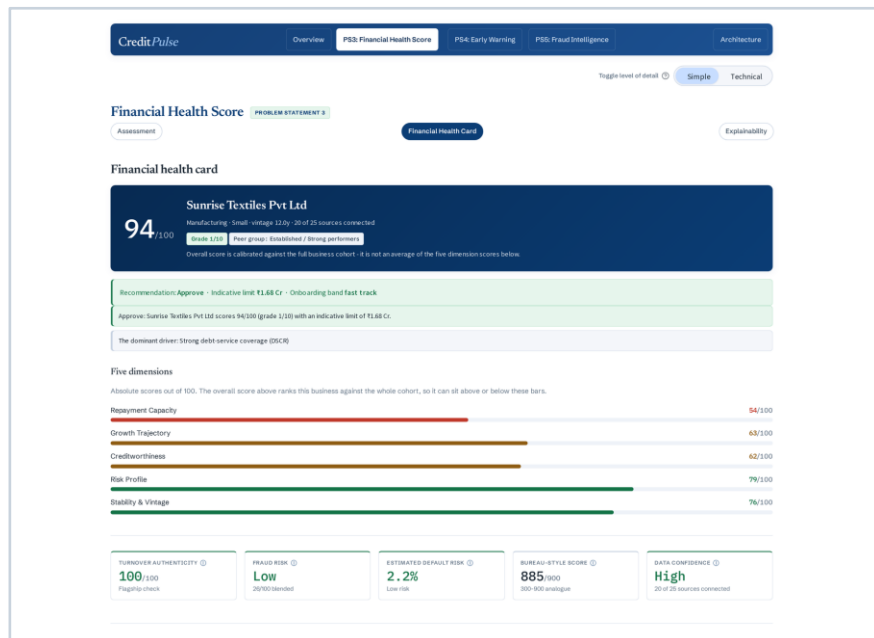
- Python 3.12; pandas + NumPy (data layer), scikit-learn (scorecard, calibration, clustering), LightGBM (monotonicity-constrained gradient boosting), SHAP (explainability)
- Streamlit multi-page product (st.navigation router); Plotly charts; Graphviz architecture diagrams
- Deterministic synthetic-data generators calibrated to public India MSME statistics (RBI, SIDBI-CRISIL, Udyam); fixed seeds, identical output on every run
- Docker single-container deploy (Cloud Run today, portable to the bank's AWS / ACC sandbox at stage 2); GitHub Actions CI regenerates data, runs the full test suite and eval harness, and builds the image on every push
- pytest: 296 tests covering determinism, monotonicity, leakage guards, UI smoke sweeps and track isolation
- No runtime LLM and no black-box service calls: every line of text on screen is assembled from the underlying computation, so the product can be audited end to end

Estimated implementation cost (optional)

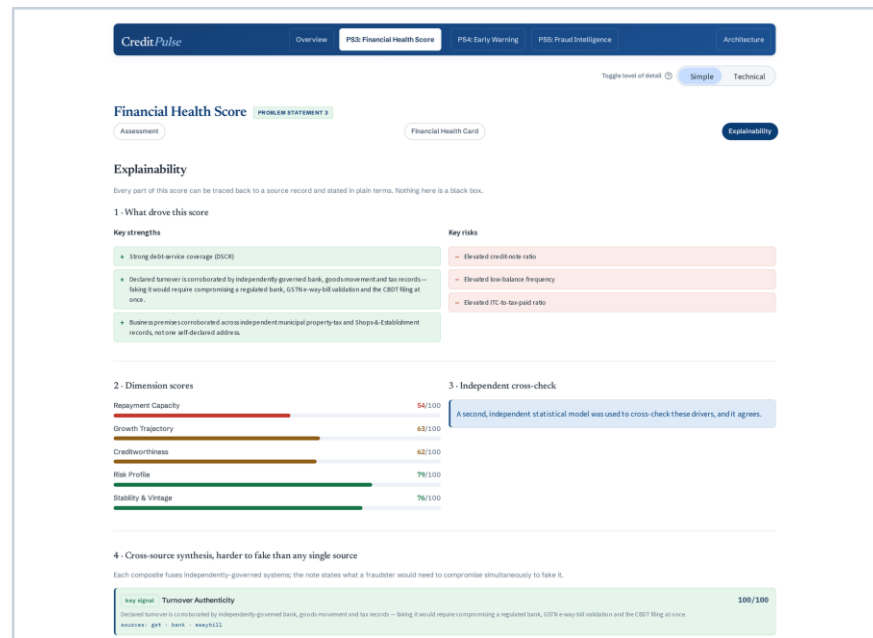
Indicative figures, stage-1 prototype scope.

- Open-source stack end to end: no licence fees, no per-score charges
- One small container (2 vCPU / 2 GB, scale-to-zero) serves all three problem statements; indicative cloud cost well under Rs 5,000 per month at demo traffic
- Pilot-phase cost is engineering, not infrastructure: AA / ULI adapter integration, DPDP consent flows, security review, and model recalibration on bank data
- Scoring is stateless and batch-friendly: portfolio-scale scoring adds compute linearly, with no specialised hardware

Snapshots of the prototype



Financial Health Card: grade, five dimensions, recommendation, indicative limit and the turnover-authenticity check



Explainability: reason codes and drivers behind every score

Prototype Performance report/Benchmarking

Eval harness scorecard - synthetic, leakage-resistant holdout (406 entities, 93 features, 286 train / 120 test)

- Default discrimination: monotonic LightGBM AUC 0.820, Gini 0.639, KS 0.572; deterministic composite AUC 0.829; WOE scorecard AUC 0.770
- Probability calibration: calibrated LightGBM Brier 0.105, ECE 0.050 - honest default probabilities, not just rankings
- Turnover-authenticity / fraud ensemble: AUC 0.949, KS 0.856 - the cross-check holds up
- Stability: max PSI 0.108 between train and holdout; demo archetypes are pinned to the training side so they can never inflate test metrics
- Reproducible with one command (make eval); full 296-test suite green

All metrics are measured on synthetic data calibrated to public India MSME statistics. Real-default backtesting and recalibration is the pilot step.

Additional Details/Future Development (if any)

- Stage 2 (sandbox): refit on IDBI mock data; replace synthetic generators with live AA / ULI / OCEN and sandbox APIs; API-first score endpoint for LOS integration
- Officer workflow: override capture, portfolio dashboards, and a model-risk documentation pack aligned to RBI digital-lending and model-governance expectations
- Real-default backtesting and recalibration; champion-challenger against the incumbent underwriting process
- The platform already monitors the booked loan (Problem Statement 4) and protects the payment rails (Problem Statement 5): one lens across the whole credit lifecycle

Provide links to your:

- GitHub Public Repository
- Demo Video Link (3 Minutes)
- Final Product Link

GitHub Public Repository:

<https://github.com/lambdacc/idbi>

Demo Video Link (3 Minutes):

<https://youtu.be/3cRJ37wFTsQ>

Final Product Link:

<https://creditpulse-66armtf3tq-el.a.run.app/>

Note: On the initial load of the demo deployment, the application may take a few seconds to appear. This is due to an on-demand startup mechanism used to minimize cloud hosting costs during the evaluation period. Once the application has loaded, please select the appropriate Problem Statement tab.

Data honesty and guardrails

- All data in the prototype is synthetic: generated, clearly labelled on screen, and calibrated to public India MSME statistics. Ground-truth labels exist for evaluation only and are never read at score time.
- Built to resist gaming: monotonicity constraints (a worsening risk factor can never improve a score), leakage-resistant holdouts, and determinism tests, all enforced in the 296-test suite
- Every user-visible number is computed from the in-repo eval harnesses at build time, or is labelled illustrative
- Human-in-the-loop: the system recommends, the officer or analyst decides, and every decision leaves an audit trail
- What we do not claim: real-world accuracy. That is measured at pilot, on bank data, after recalibration.



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Thank you!